

Pandas Analyzer & Data Visualization

Professional Project Documentation

This project is a comprehensive Sales Data Analysis and Visualization system developed using Python, Pandas, NumPy, Matplotlib, and Seaborn. The project demonstrates data exploration, cleaning, statistical analysis, aggregation, pivot table creation, and multiple visualization techniques using a real-world sales dataset.

Key Features

- Dataset Loading and Exploration
- Data Cleaning and Preprocessing
- NumPy Array Operations
- Search, Sort, and Filter Operations
- Aggregate Functions and Statistical Analysis
- Pivot Table Generation
- Matplotlib Visualizations (Bar, Line, Scatter, Pie, Histogram, Stack Plot)
- Seaborn Visualizations (Heatmap, Box Plot)
- Menu-Driven Interface using OOP Concepts

Dataset

Dataset Used: 1000 Sales Records.csv. Attributes include Region, Country, Item Type, Sales Channel, Order Date, Units Sold, Unit Price, Unit Cost, Total Revenue, Total Cost, and Total Profit.

Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, and Jupyter Notebook.

Project Outcomes

- Revenue and Profit Analysis
- Regional Sales Comparison
- Business Trend Identification
- Statistical Insight Generation
- Professional Visualization Reports

Conclusion

The project successfully demonstrates practical implementation of data analysis and visualization techniques using Python. It provides valuable business insights and serves as a strong academic project for Data Analysis coursework.

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